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SEAT No. _____

Total No. of printed pages: 02

SARDAR PATEL UNIVERSITY

M.Sc. (Information Technology)

Semester – III External Examinations

PS03EINT02 (Data Mining and Data Warehousing)

13th November, 2017 (Monday)

Time: 10:00 AM to 01:00 PM

Max. Marks: 70

Q-1 Choose most appropriate option for the following: [08]

1. In archiving, _____ data is removed from the system in a format that allows it to be quickly restored if required.
A. Newer
B. Older
C. A and B both
D. None of these
2. _____ process is the system process that manages the queries and speeds up them by directing queries to the most effective data source.
A. Data recovery
B. Query management
C. Archiving
D. None of these
3. The _____ is the system component that performs all the operations necessary to support the warehouse management process.
A. Load manager
B. Query manager
C. Warehouse manager
D. None of these
4. _____ contains textual descriptions.
A. Fact information
B. Reference information
C. A and B both
D. None of these
5. _____ is where a single logical entity is partitioned into multiple sub-entities.
A. Aggregation
B. Partitioning
C. Association
D. None of these
6. _____ is to find patterns in observations.
A. Observation
B. Analysis
C. Theory
D. Prediction
7. A _____ is a backup that is taken while the database is completely shut down.
A. Hot backup
B. Online backup
C. A and B both
D. None of these

8. _____ is the information that is used to analyze the elemental transaction.
- A. Dimension data
 - B. Fact data
 - C. A and B
 - D. None of these

Q-2 Write answers in short: (ANY SEVEN) [14]

- a. What is data mining? Explain it in brief.
- b. What is data warehousing? Explain it in brief.
- c. Explain importance of backup process within a data warehouse.
- d. Discuss metadata taking suitable example.
- e. Explain summary information in brief.
- f. Explain starflake schemas.
- g. Explain fact table design in brief.
- h. Define: operational system and mission critical system
- i. What is SLA? Explain it in brief.

Q-3 A Explain project planning and management life cycle. [06]
B Discuss various application areas of data mining. [06]

OR

B Explain self-learning computer systems. [06]

Q-4 A Explain extract and load process within a data warehouse. [06]
B Write a short note on query manager. [06]

OR

B Explain clean and transform data process within a data warehouse. [06]

Q-5 A Explain data marting taking suitable example. [06]
B Write a short note on vertical partitioning. [06]

OR

B Explain data aggregation taking suitable examples. [06]

Q-6 A Discuss security requirements for data warehouse. [06]
B Explain client hardware requirement of data warehouse environment. [06]

OR

B Write a short note on horizontal partitioning. [06]

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